0363-5465/82/1002-0103$02.00/0
THE AMERICAN JOURNAL OF SPORTS MEDICINE, Vol. 10, No. 2
© 1982 American Orthopaedic Society for Sports Medicine

Primary surgical treatment of anterior cruciate ligament lesions*

JOHN L. MARSHALL,† DVM, MD, FACS, RUSSELL F. WARREN,‡ MD, FACS, AND THOMAS L. WICKIEWICZ,§‖ MD

From the Sports Medicine Department, The Hospital for Special Surgery, affiliated with The New York Hospital, Cornell University Medical College

ABSTRACT

Seventy patients who underwent primary repair of the anterior cruciate ligament are reviewed in follow-up. The surgical technique consisted of multiple loop-varying depth sutures in both cruciate stumps in 61 patients. Nine patients had additional fascial augmentation done primarily. Follow-up was from 12 to 90 months, the average being 29 months. All patients were scored on a 50-point normal knee score sheet. Average score at follow-up was 42.7. No patient was bothered by giving way, which is defined as an unpredictable instability. No patient has required subsequent meniscal surgery. Ninety-three percent of the patients were active in sports. Anterior drawer sign at 90° showed increased excursion compared to the unaffected leg in all patients. However, Lachman's test showed a firm end point to excursion in all knees tested, and, in addition, 52% were rated as normal compared to the unaffected knee. We conclude that primary repairs of mid-substance tears are technically possible and recommended in an athlete.

Although the anterior cruciate ligament's importance for normal knee function is generally accepted, a primary repair which yields consistently good results is still controversial.

Early work by O'Donoghue[1,2] in dogs showed ligament healing after surgical transection. Recent work of Cabaud et al.[3] in dogs and monkeys confirmed this, and also provided information on the relative strength of these repairs.

Clinical studies in humans have not been encouraging. Liljedahl et al.[4] reported excellent results in 1965; however, the series was small (25/33) and the follow-up short (6–18 months). Eriksson[5] recommends primary repair and reported good results in 1976. Feagin's long-term results in his five year follow-up[6] were poor, with instability recorded in 94%, and significant reinjury in 17/32 patients. His technique of repair was a figure 8 suture of either chromic or mersiline brought out through bone. More recently, Feagin recommends augmentation with a fascial graft in the primary repairs.[7]

The cruciate insufficient knee in an athletic individual is destined for problems.[8,9] This syndrome of cruciate insufficiency is manifested by an unpredictable instability, (termed "giving way"), and is perceived by the patient as one bone slipping on the other. The knee is painful and often develops subsequent swelling. Both the onset and the severity of the giving way is quite variable, ranging from a few months to several years post-injury. Generally, it is related to the level of stressful sports activity of the individual and is precipitated particularly by sports that require jumping and/or cutting. Instability is followed by meniscal degeneration, frequent x-ray findings of tibial spine, and medial joint line spurring, and, in some instances, premature degenerative osteoarthritis.

Successful primary repair would obviate these problems. The authors feel that certain anatomic and biologic principles must be adhered to in performing a primary repair to obtain both excellent and reproducible results.

MATERIALS AND METHODS

The surgical technique[10] consists of varying depth, multiple-loop sutures of O. Tevdek (Deknatel, New York, New York) in the cruciate stumps. Care is taken to reconstitute the normal anatomy of the anterior cruciate ligament. The an-

*Presented at the interim meeting of the American Orthopedic Society for Sports Medicine, Atlanta, Georgia, February 5–7, 1980.

† Associate Professor of Surgery (Orthopedics) and Assistant Professor of Anatomy, Cornell University Medical College (deceased).

‡ Chief, The Sports Medicine Service, The Hospital for Special Surgery; Assistant Professor of Surgery (Orthopedics), Cornell University Medical College.

§ Orthopedic Surgical Fellow, University of California at Los Angeles.

‖Address correspondence to: Thomas L. Wickiewicz, MD, The Hospital for Special Surgery, 535 E. 70th Street, New York, New York 10021.

teromedial band is attached posteriorly and superiorly on the lateral femoral condyle via a drill hole or brought out over the top of the lateral condyle. Then, through a second drill hole, the posterolateral bulk is attached slightly inferior and anterior to this site. Careful placement of the drill holes is necessary to achieve a broad attachment, thus affording normal length changes in the ligament as the knee passes through a range of motion. The fat-pad, ligamentum mucosum, and remaining synovial sheath are sutured into the repaired ligament as a source of blood supply.[11]

In addition, in those infrequent cases where ligament destruction precluded an adequate repair, a fascial reinforcement was performed using a tube of fascia lata, based distally, and brought over the top of the lateral femoral condyle. This was attached anteriorly on the tibial shaft and, in addition, was sutured side-to-side with the repaired ligament.

Postoperative care consisted of a cylinder cast for 6 weeks, with gradual progression to full weight bearing. This was followed by bracing for an additional 2 months while range of motion and muscle strengthening exercises were begun. In general, running was not begun until 4–5 months postoperatively.

This series consisted of 70 consecutive primary repairs of the anterior cruciate ligament. The patients were under the care of the two senior authors.

Our patient population included 50 males, average age 26.1 years, and 20 females, average age 27.8 years, with an overall age range of 13–50 years old. Almost all injuries occurred during some form of sports participation, with patients varying from the infrequent recreational athlete to the professional athlete.

The average duration from time of injury to time of surgery was 5.3 days with a range of 1 to 26 days.

The diagnosis was based on history, physical examination, arthrography, examination under anesthesia and, rarely, arthroscopy. The anterior drawer sign was performed in all patients, and although the amount of excursion varied, the absence of a firm end point was noted. Lachman's test or a drawer test in 20° of flexion was much more sensitive, and our most reliable test. As in the anterior drawer sign, the absence of a firm end point constituted a positive test. These tests are easily performed, even without anesthesia, in the acutely injured knee. The pivot shift was not recorded in the early part of this series, but it was tested in 40 patients and known to be positive in 38.

The ligament tear location, as observed in arthrotomy, was proximal in 13 knees, distal, with an associated fleck of bone in one, and a mid-substance, mop-end tear in 56.

TABLE 1
Pattern of associated injuries in the 70 patients

	Knees
Isolated ACL	20
ACL + MCL	38
ACL + PCL	6
ACL + lateral side (LCL, ITB, biceps)	2
ACL + tibial shaft fracture	1
Meniscal injuries associated with the above lesions	26

Twenty knees were considered to be clinically isolated cruciate injuries. Associated injuries in the remaining 50 knees are listed in Table 1.

Primary repair alone (multiple loop suture technique) was carried out for 61 knees. In the remaining nine knees, the fascia lata reinforcement augmented primary repair. Associated injuries were repaired in addition to the cruciate ligament.

RESULTS

All patients were re-examined at follow-up by one of the authors. Knees were scored using our 50-point Knee Score Sheet,[12] in which 50 points are awarded to a normal, asymptomatic knee. The absence of giving way, return to sports, and static stability on clinical testing are heavily weighted.

Minimum follow-up time was 12 months, and the range of follow-up in this series was from 12 to 90 months, with an average of 29 months.

Twenty-five patients complained of pain in the operated knee, not present before injury, and often diagnosed at follow-up as patellar in origin. Of these patients, 50% had measurable thigh atrophy. Three patients complained of swelling, intermittent in nature, and not associated with trauma. All three were evaluated at only 12 months postoperatively. Forty-four were bothered by clicks and/or crepitus not present preoperatively, and also by residual numbness in the area of the infrapatellar branch of the saphenous nerve.

All patients felt their knees were stable postoperatively and no one complained of the "giving way" classically associated with cruciate insufficiency. In addition, no patient has required a second operative procedure for meniscal degeneration.

The ability to return to sports participation was carefully assessed in all patients. Forty-eight (69%) noted no restriction of activity and were participating fully at their preinjury level of sports competition at an average follow-up of 32 months.

Seventeen patients (24%) felt some restriction in their activity but were active in some form of sports, such as tennis or paddle ball. Inability to return to the sport in which they were injured was the prime reason for classification in this group. It should be noted that these patients had a shorter follow-up time of 18 months. Four patients (5.7%) participated only in the mildest of sports activity, such as jogging, postinjury, and had an average follow-up of 19 months. In two of these, there was associated posterior cruciate injury. One patient, who had an associated tibial shaft fracture, was not yet rehabilitated, and was not participating in any form of sports activity, although clinical testing showed his knee was stable.

Eleven patients exhibited tenderness on physical examination, most often peripatellar in location, and two patients, at a 12 month follow-up, had mild effusions. Muscle strength and thigh size were normal in 44 patients (63%). The remaining 37% exhibited residual muscle atrophy in the operated leg.

Range of motion was full in 20 patients (29%). Three patients showed loss of the terminal few degrees of extension, and 47 (67%) had loss of terminal flexion, averaging 10°.

Three clinical tests were used to assess anterior cruciate ligament integrity: the anterior drawer sign in neutral, the Lachman test, and the pivot shift test.

The anterior drawer test was rated on a scale of 0 to 5 points, with 5 as normal, 4 as a mild displacement, 3 as a moderate displacement, and 2 as severe in neutral rotation. No patient exhibited a drawer sign rated as normal or equal to the other knee. Thirty-five (50%) were rated as 4, 34 (48%) were rated as 3, and one was assessed as a 2. A zero score was given if a knee's anterior drawer increased with rotation, (e.g., positive Slocum test). No patient in the series had this. A score of one was not used in this classification.

Results on the Lachman test were more impressive. Thirty-four (47%) were rated as normal, or equal to the unaffected knee. Thirty-two (45%) showed a slight increase in excursion compared to the normal knee. In four patients, the test was not recorded. In all 66 patients (92%) in whom the test was performed, a firm end point to excursion, not present preoperatively, was clearly demonstrable.

The pivot shift test postsurgically was recorded in 63 patients. It was negative in 47 (75%) and positive in 16 (25%). In the 38 patients in whom it was known to be positive preoperatively, it was negative in 29 postoperatively, and remained positive in 9 (Table 2).

TABLE 2
Pivot shift[a]

	Whole group (70 pts.)		Isolated group (20 pts.)	
	Pre-op	Post-op	Pre-op	Post-op
Positive	38	16	16	3
Negative	2	47	1	17
Not recorded	30	7	3	—

[a] Pre- and postoperative results for the whole series and for the isolated subgroup. For those patients in whom the shift was known to be positive preoperatively, it was eliminated in 76%.

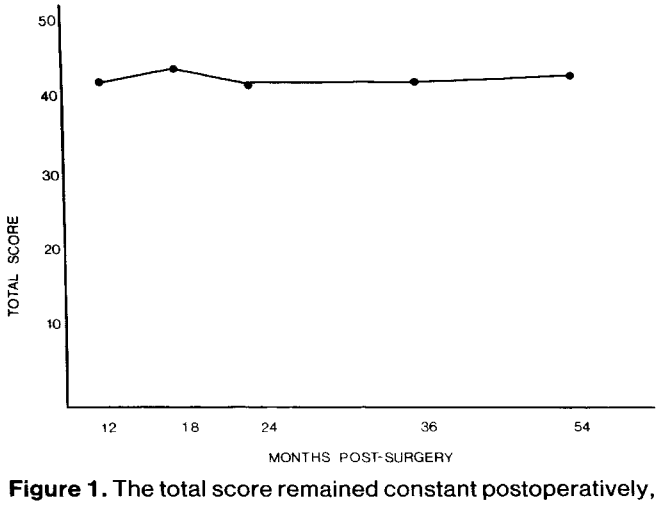


Figure 1. The total score remained constant postoperatively, even past 5 years.

The total score, based on a 50-point system, averaged 42.7. It was slightly higher (44) for the subgroup of clinically isolated cruciate injuries (20 patients). Also, this score showed no deterioration with time postsurgically. For 40 patients with a follow-up of 1 to 2 years, the average score was 42.5. Thirty patients in the 3- to 8-year follow-up range averaged 42 (Fig. 1).

Complications of surgery included one superficial wound infection (*Staphylococcus aureus*) which required a second procedure for wound debridement, but did not compromise the repair, and two patients required manipulations under anesthesia as a second procedure because of poor range of motion.

There was one reinjury, in a professional football player at 1 year postsurgery. His initial surgery included a fascia lata reinforcement. At the time of the second surgery, a mop end tear of the ligament was found, and the stumps appeared robust with acute hemorrhage present throughout the ligament on pathological review.

DISCUSSION

Most of the criticism regarding primary repair of the anterior cruciate ligament centers on the inability to technically repair midportion mop end tears and, if repaired, the unlikeliness of healing. We have noted by using the multiple loop suture technique that primary repair is feasible and the results reproducible. Care must be taken at the time of surgery to both recreate the anatomy and encourage the blood supply by reattaching the synovium and the fat pad.

Our criterion for success after surgical repair is primarily a functional one. Our major goal has been to allow the athlete to return to stressful sports actively without experiencing instability. Objective criteria are not overlooked, however. The presence of a firm end point on the Lachman test confirms the integrity of the repaired ligament. This test gives no indication of the strength of the repair, only that the repaired ligament remains intact.

For evaluating our patients, we have been concerned about the persistent drawer at 90° in most patients. It should be noted that this was severe in only two patients and was accompanied by a firm end point in 68 patients. In addition, no patient has, to date, noted the symptom of giving way or required surgery for subsequent meniscal degeneration. Thus it appears the repair is functional and provides static support. Further evidence for this is the fact that the Lachman test was normal in 52% with only a slight jog in 48% and a firm end point in all. This would indicate better stability near extension, the position where the athlete tends to note pivoting and have subluxing episodes.

At follow-up, the pivot shift was positive in 25% of the 63 patients tested. Recently, we have felt that the pivot shift should be evaluated on a scale of 1 to 3, ranging from a mild subtle shift to a gross shift. Most of the patients with a persistant shift had the milder form consisting of lateral crepitation in testing without a complete slip being noted.

The functional return to sports has been excellent. Sixty-five patients (93%) are actively involved in sports and 48

(68%) are participating freely at their preinjury level. Of these 48, 12 were professional or world class athletes. Bracing has not been required for any sport. It is also important to note that the level of sports participation has improved with time, rather than deteriorated. The group of patients rated as having a full return to sports averaged 32 months post-surgery as compared to 18 months for the group with a partial return. In addition, of those patients whose follow-up was a minimum of 2 years (30), averaging 4.5 years, there has been no deterioration of the results with time. This is in contrast to the results of Feagin et al. who report, at 2- and 5-year follow-up, meniscal deterioration resulting in meniscectomy in 16/64 patients and complaints of instability in 94%.

We feel that the absence of knee instability with stressful sports in all patients and the fact that none have required a second operative procedure for meniscal degeneration both offer evidence of the success of the repair to date. It is our impression that primary surgical treatment of the torn anterior cruciate ligament is indicated in young athletically inclined individuals and can be performed with reasonable assurance of success. Augmentation may be required but has not been a routine component of primary repair.

REFERENCES

1. O'Donoghue D, Frank G, Jeter G, et al: Repair and reconstruction of anterior cruciate ligament in dogs. *J Bone Joint Surg 53A:* 710–718, 1971
2. O'Donoghue D, Rockwood C, Frank G, et al: Repair of the anterior cruciate ligament in dogs. *J Bone Joint Surg 48A:* 503–519, 1966
3. Cabaud HE, Rodkey W, Feagin J: Experimental studies of acute anterior cruciate ligament injury and repair. *Am J Sports Med 7:* 18–22, 1979
4. Liljedahl SO, Lindvall N, Wetterfors J: Early diagnosis and treatment of acute ruptures of the anterior cruciate ligament. *J Bone Joint Surg 47A:* 1503–1513, 1965
5. Eriksson E: Sports injuries of the knee ligaments: Their diagnosis, treatment, rehabilitation and prevention. *Med Sci Sports 8:* 133–144, 1976
6. Feagin J, Curl W: Isolated tear of the anterior cruciate ligament. Five year follow-up study. *Am J Sports Med 4:* 95–100, 1976
7. Feagin J: The syndrome of the torn anterior cruciate ligament. *Orthop Clin North Am 10:* 81–90, 1979
8. Fetto JF, Marshall JL: The natural history and diagnosis of anterior cruciate ligament insufficiency. *Clin Orthop 147:* 29–38, 1980
9. Warren RF, Marshall JL: Injuries to the anterior cruciate and medial collateral ligaments of the knee. Parts I & II. *Clin Orthop 136:* 198–211, 1978
10. Marshall JL, Warren RF, Wickiewicz TL, et al: The anterior cruciate ligament. A technique of repair and reconstruction. *Clin Orthop 143:* 97–106, 1979
11. Marshall JL, Arnoczky SP, Rubin RM, et al: Microvasculature of the cruciate ligaments. *Physician Sportsmed 7:* 87–91, 1979
12. Marshall JL, Fetto JF, Botero PM: Knee injuries: A standardized evaluation method. *Clin Orthop 123:* 115–129, 1977

COMMENTARY

Ejnar Eriksson, M.D., *Stockholm, Sweden*: I would like to thank the authors for an interesting presentation. I agree with practically everything said in the paper.

Many of you who have listened to me through the years know that in Sweden we have favored repair of the acutely torn anterior cruciate since Ivar Palmer's time in the 1930's and we have all been taught to do the procedure. The surgical technique is not all that easy, and the results probably depend on how accurately the procedure is performed. I would like to point out how essential it is that you make the diagnosis immediately, because the earlier you operate, the easier it is to do the repair.

I would like to stress that physicians should consider doing arthroscopy on most of the athletes that come in after a knee injury with an acute effusion in the knee joint. I strongly feel that if you do so, you will diagnose a number of torn anterior cruciates which you otherwise would not recognize until later on, and then have to reconstruct.

Fortunately, the ligament does stretch out a little before it ruptures. Therefore, it is usually long enough to reach back to its posterior insertion. It has to be brought back into its anatomical position. If the rupture has taken place in the midsubstance, we reinforce the acute repair with a reconstruction, using the medial third of the patellar tendon.

Authors' Reply: There are two comments I would like to make. One is that, of the fascial reinforcements reported in this series, a majority were performed earlier in this series, and as the techniques of placement of the multiple loop sutures improved, we found there was less need for reinforcement.

The only additional comment I would like to make is that, often, when you first open the knee and look for that proximal fragment, it looks as if there is nothing there you can get a suture into, but if you spend a little time and carefully dissect away the synovium without sacrificing it, you will—nine times out of ten—find enough material of that proximal fragment that can be sutured.

EDITORIAL COMMENTARY

Victor H. Frankel, MD, *Hospital for Joint Diseases, New York, New York*: The senior author, whose untimely death closed all too soon a brilliant career in sports medicine, pointed out very early in his work on experimental division of the anterior cruciate ligament why this is not only the cruciate, but the "crucial," ligament. Experimental work and careful clinical observation utilizing an intricate scoring technique have led the authors to develop this series in which early primary repair of the anterior cruciate ligament has been performed and assessed.

The follow-ups, which averaged approximately 2½ years, had a knee score of 42.7. This compares very favorably to a knee score of around 30 for the untreated group over the same time interval. Certainly, primary repair of the anterior cruciate ligament and associated ligamentous injuries has resulted in a better objective knee score. Other criteria for this series have included return to sports participation, Lachman's test, and giving way.

Pivot shift was also used in a portion of the group. The operated group was benefited by surgery, utilizing these criteria. I wonder, however, if participation in sports can not be discounted, since many motivated athletes will overlook some complaints, or are so well conditioned that they can overcome ligamentous instability with muscle function.

Lachman's test and anterior drawer test are basically subjective tests of the examiner. Modern techniques using devices such as developed by Robert Johnson of Vermont, Jack Kennedy in London, or Keith Markolf in Los Angeles can produce the very objective information as to ligamentous stability. Giving way, of course is a subjective symptom which may be controlled, in some, by the amount of loading they will put on their knee, or by muscle conditioning. The pivot shift test can be an objective test, but the authors state that they are trying to grade this from one to three to make it more useful.

The authors describe a multiple, loop-varying depth technique for suturing the cruciate ligaments, and carefully approximate the normal anatomy utilizing either two holes in the femoral condyle, or one hole and over-the-top technique. Of the 70 ruptured cruciates, 13 were at the proximal end, and these could be drawn back upwards by sutures. Fifty-six were mid-ligamentous ruptures, and it would seem that the sutures were pulled upward to the femoral attachment, but that the torn ends remained in the body of the ligament where healing would have to take place. Adjacent structures were sutured into the repaired ligament as a source of blood supply.

Rupture of the ligament occurs when the ligament passes the elastic and plastic zones in its loading curve until failure results. Even though the ligament ends have been sewn together, ligamentous fibers have been damaged by loading through the plastic zone. I wonder if the end result of the repair is a ligament or modified scar tissue. The authors do not report any direct observation on the ligaments through arthroscopy to determine the character of the ligament.

In summary, the authors present an operation for acute rupture of the anterior cruciate ligament which depends upon a meticulous and skillful repair. This technique seems to offer the injured a chance to achieve a stable knee which will allow him to return to some level of sports participation. Whether or not this early repair will prevent late degenerative arthritis is still open to question.